

# An Introduction to the Bootstrap

## 1 The Bootstrap

The bootstrap was introduced by [Efron \[1979\]](#) and is considered as one of the most important ideas in modern statistics.

The following slides are adapted from Hastie's lecture notes and provide a concrete example illustrating the bootstrap principle.

## A simple example

- Suppose that we wish to invest a fixed sum of money in two financial assets that yield returns of  $X$  and  $Y$ , respectively, where  $X$  and  $Y$  are random quantities.
- We will invest a fraction  $\alpha$  of our money in  $X$ , and will invest the remaining  $1 - \alpha$  in  $Y$ .
- We wish to choose  $\alpha$  to minimize the total risk, or variance, of our investment. In other words, we want to minimize  $\text{Var}(\alpha X + (1 - \alpha)Y)$ .
- One can show that the value that minimizes the risk is given by

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}},$$

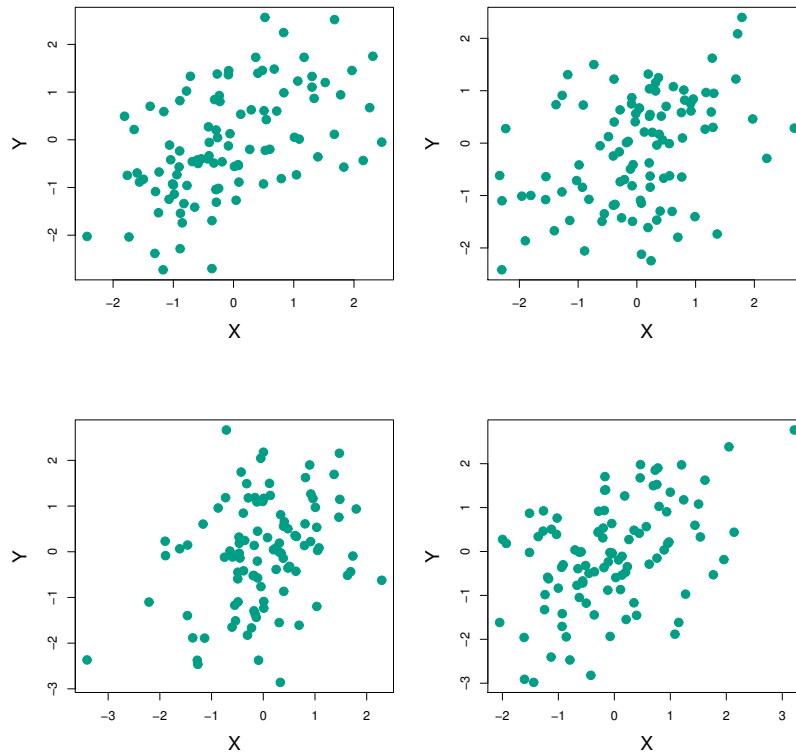
where  $\sigma_X^2 = \text{Var}(X)$ ,  $\sigma_Y^2 = \text{Var}(Y)$ , and  $\sigma_{XY} = \text{Cov}(X, Y)$ .

## Example continued

- But the values of  $\sigma_X^2$ ,  $\sigma_Y^2$ , and  $\sigma_{XY}$  are unknown.
- We can compute estimates for these quantities,  $\hat{\sigma}_X^2$ ,  $\hat{\sigma}_Y^2$ , and  $\hat{\sigma}_{XY}$ , using a data set that contains measurements for  $X$  and  $Y$ .
- We can then estimate the value of  $\alpha$  that minimizes the variance of our investment using

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}.$$

## Example continued



*Each panel displays 100 simulated returns for investments  $X$  and  $Y$ . From left to right and top to bottom, the resulting estimates for  $\alpha$  are 0.576, 0.532, 0.657, and 0.651.*

## Example continued

- To estimate the standard deviation of  $\hat{\alpha}$ , we repeated the process of simulating 100 paired observations of  $X$  and  $Y$ , and estimating  $\alpha$  1,000 times.
- We thereby obtained 1,000 estimates for  $\alpha$ , which we can call  $\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_{1000}$ .
- The left-hand panel of the Figure on slide 29 displays a histogram of the resulting estimates.
- For these simulations the parameters were set to  $\sigma_X^2 = 1, \sigma_Y^2 = 1.25$ , and  $\sigma_{XY} = 0.5$ , and so we know that the true value of  $\alpha$  is 0.6 (indicated by the red line).

## Example continued

- The mean over all 1,000 estimates for  $\alpha$  is

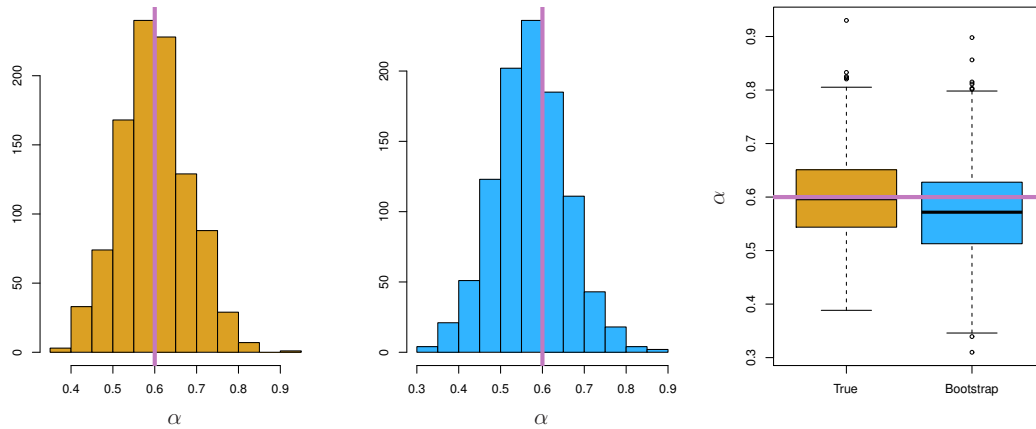
$$\bar{\alpha} = \frac{1}{1000} \sum_{r=1}^{1000} \hat{\alpha}_r = 0.5996,$$

very close to  $\alpha = 0.6$ , and the standard deviation of the estimates is

$$\sqrt{\frac{1}{1000 - 1} \sum_{r=1}^{1000} (\hat{\alpha}_r - \bar{\alpha})^2} = 0.083.$$

- This gives us a very good idea of the accuracy of  $\hat{\alpha}$ :  $\text{SE}(\hat{\alpha}) \approx 0.083$ .
- So roughly speaking, for a random sample from the population, we would expect  $\hat{\alpha}$  to differ from  $\alpha$  by approximately 0.08, on average.

## Results



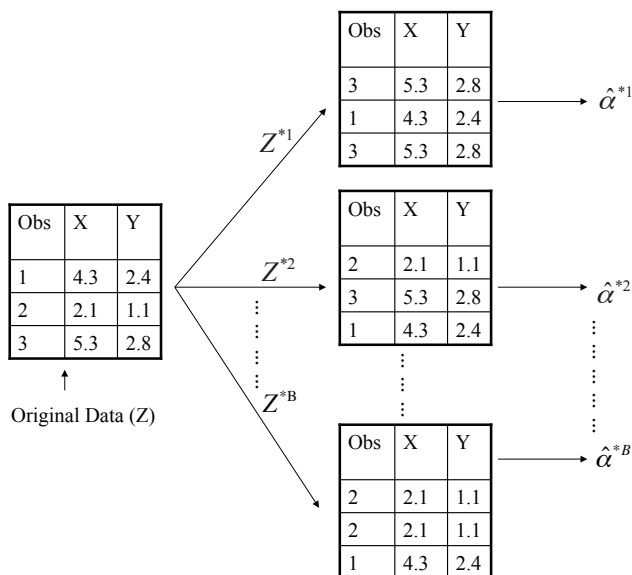
*Left:* A histogram of the estimates of  $\alpha$  obtained by generating 1,000 simulated data sets from the true population. *Center:* A histogram of the estimates of  $\alpha$  obtained from 1,000 bootstrap samples from a single data set. *Right:* The estimates of  $\alpha$  displayed in the left and center panels are shown as boxplots. In each panel, the pink line indicates the true value of  $\alpha$ .

## Now back to the real world

- The procedure outlined above cannot be applied, because for real data we cannot generate new samples from the original population.
- However, the bootstrap approach allows us to use a computer to mimic the process of obtaining new data sets, so that we can estimate the variability of our estimate without generating additional samples.
- Rather than repeatedly obtaining independent data sets from the population, we instead obtain distinct data sets by repeatedly sampling observations from the original data set *with replacement*.
- Each of these “bootstrap data sets” is created by sampling *with replacement*, and is the *same size* as our original dataset. As a result some observations may appear more than once in a given bootstrap data set and some not at all.



## Example with just 3 observations



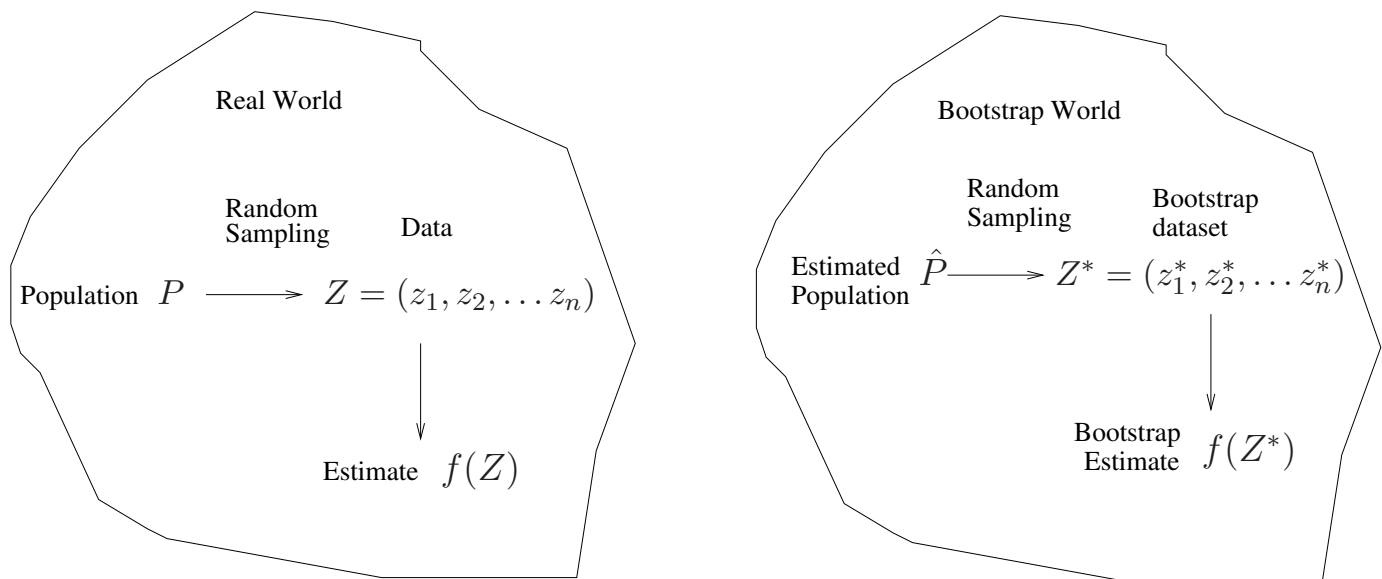
A graphical illustration of the bootstrap approach on a small sample containing  $n = 3$  observations. Each bootstrap data set contains  $n$  observations, sampled with replacement from the original data set. Each bootstrap data set is used to obtain an estimate of  $\alpha$

- Denoting the first bootstrap data set by  $Z^{*1}$ , we use  $Z^{*1}$  to produce a new bootstrap estimate for  $\alpha$ , which we call  $\hat{\alpha}^{*1}$
- This procedure is repeated  $B$  times for some large value of  $B$  (say 100 or 1000), in order to produce  $B$  different bootstrap data sets,  $Z^{*1}, Z^{*2}, \dots, Z^{*B}$ , and  $B$  corresponding  $\alpha$  estimates,  $\hat{\alpha}^{*1}, \hat{\alpha}^{*2}, \dots, \hat{\alpha}^{*B}$ .
- We estimate the standard error of these bootstrap estimates using the formula

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1} \sum_{r=1}^B (\hat{\alpha}^{*r} - \bar{\hat{\alpha}}^*)^2}.$$

- This serves as an estimate of the standard error of  $\hat{\alpha}$  estimated from the original data set. See center and right panels of Figure on slide 29. Bootstrap results are in blue. For this example  $SE_B(\hat{\alpha}) = 0.087$ .

## A general picture for the bootstrap



## 2 A Functional View of the Bootstrap

Suppose

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P, \quad (1)$$

where  $P$  is an unknown probability distribution.

Let

$$T_n = T(X_1, \dots, X_n) \quad (2)$$

denote a statistic of interest. Examples of such statistics include the sample mean, the sample median, regression and correlation coefficients, and the portfolio estimator discussed in the preceding slides. In many applications, our goal is not merely to compute the value of  $T_n$ , but also to quantify its uncertainty.

For example, we may wish to know the variance of  $T_n$ ,

$$\text{Var}_P(T_n). \quad (3)$$

The subscript  $P$  emphasizes that this variance is computed under repeated sampling from the population distribution  $P$ . For a fixed sample size  $n$ , the quantity  $\text{Var}_P(T_n)$  depends only on the underlying distribution  $P$ .

This suggests viewing the variance as a *functional* of the population distribution:

$$f(P) = \text{Var}_P(T_n). \quad (4)$$

More generally, many quantities of statistical interest can be expressed as functionals of  $P$ . Important examples are the bias and mean squared error of an estimator, quantiles of the sampling distribution of  $T_n$ .

If the distribution  $P$  were known, then estimating  $f(P)$  would be straightforward. We could repeatedly generate independent samples of size  $n$  from  $P$ , compute the statistic  $T_n$  on each sample, and use the resulting Monte Carlo sample to estimate  $f(P)$ .

The difficulty, of course, is that  $P$  is unknown.

The bootstrap proceeds by replacing the unknown distribution  $P$  with an estimate of  $P$ , namely the empirical distribution

$$\hat{P}_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}, \quad (5)$$

which places mass  $1/n$  at each observed data point.

The bootstrap estimate of  $f(P)$  is then obtained by the plug-in principle:

$$f(P) \approx f(\hat{P}_n). \quad (6)$$

In other words, we behave as if the empirical distribution were the true population distribution. We repeatedly sample from  $\hat{P}_n$ , compute the statistic on each resampled data set, and use the resulting variation to estimate the uncertainty of  $T_n$ .

From this viewpoint, the bootstrap is simply a method for approximating an unknown functional  $f(P)$  by the plug-in estimator  $f(\hat{P}_n)$ .

## 2.1 The Empirical Distribution

The observed sample naturally gives rise to the *empirical distribution*

$$\hat{P}_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}, \quad (7)$$

where  $\delta_x$  denotes a point mass at the value  $x$ .

The empirical distribution assigns probability  $1/n$  to each observed data point  $X_1, \dots, X_n$ . It may be viewed as an estimate of the unknown population distribution  $P$ .

A key observation is that sampling independently from  $\hat{P}_n$  is equivalent to drawing observations from the original dataset  $\{X_1, \dots, X_n\}$  with replacement. Indeed, each draw selects one of the observed data points with equal probability  $1/n$ .

This simple fact is the foundation of the bootstrap. Since the true distribution  $P$  is unknown, we replace it by the empirical distribution  $\hat{P}_n$  and use repeated samples from  $\hat{P}_n$  to mimic repeated sampling from the population.

## 2.2 The Bootstrap Principle

The central idea of the bootstrap is the *plug-in principle*. Suppose our quantity of interest can be written as a functional

$$f(P) \quad (8)$$

of the unknown population distribution  $P$ . Since  $P$  is unavailable, we replace it by the empirical distribution  $\hat{P}_n$  and estimate  $f(P)$  by

$$f(\hat{P}_n). \quad (9)$$

For example, if

$$f(P) = \text{Var}_P(T_n), \quad (10)$$

then the bootstrap approximation is

$$\text{Var}_P(T_n) \approx \text{Var}_{\hat{P}_n}(T_n). \quad (11)$$

The quantity on the right is generally not available in closed form, but it can be approximated by Monte Carlo simulation.

The bootstrap algorithm proceeds as follows.

1. Generate a bootstrap sample

$$X_1^*, \dots, X_n^* \stackrel{\text{i.i.d.}}{\sim} \hat{P}_n. \quad (12)$$

Equivalently, draw  $n$  observations from the original dataset with replacement.

2. Compute the bootstrap statistic

$$T_n^* = T(X_1^*, \dots, X_n^*). \quad (13)$$

3. Repeat Steps 1 and 2 independently  $B$  times, producing

$$T_{n,1}^*, \dots, T_{n,B}^*. \quad (14)$$

4. Use the empirical distribution of the bootstrap statistics to estimate quantities of interest, such as their variance, standard error, bias, or quantiles.

For example, the bootstrap estimate of the variance of  $T_n$  is

$$\widehat{\text{Var}}_{\text{boot}}(T_n) = \frac{1}{B-1} \sum_{b=1}^B \left( T_{n,b}^* - \bar{T}_n^* \right)^2, \quad (15)$$

where

$$\bar{T}_n^* = \frac{1}{B} \sum_{b=1}^B T_{n,b}^*. \quad (16)$$

Thus, the bootstrap estimates the sampling variability of  $T_n$  by studying how the statistic varies across repeated samples drawn from the empirical distribution.

### 2.3 Why Should This Work?

The bootstrap is based on the idea that the empirical distribution  $\hat{P}_n$  provides a good approximation to the unknown population distribution  $P$ . If this approximation is sufficiently accurate, then we may hope that functionals of  $\hat{P}_n$  will also be close to the corresponding functionals of  $P$ .

A classical result supporting this intuition is the Glivenko–Cantelli theorem.

**Theorem 1** (Glivenko–Cantelli). *Let*

$$F(x) = P((-\infty, x]) \quad (17)$$

*denote the population distribution function, and let*

$$F_n(x) = \hat{P}_n((-\infty, x]) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq x\} \quad (18)$$

*denote the empirical distribution function.*

*Then*

$$\sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \longrightarrow 0 \quad (19)$$

*almost surely as  $n \rightarrow \infty$ .*

Thus, as the sample size increases, the empirical distribution converges uniformly to the true distribution. In a precise sense,  $\hat{P}_n$  becomes an increasingly accurate approximation to  $P$ .

It is therefore natural to expect that if a functional  $f$  depends continuously on the underlying distribution, then

$$\hat{P}_n \approx P \quad \implies \quad f(\hat{P}_n) \approx f(P). \quad (20)$$

The bootstrap exploits exactly this idea. It replaces the unknown quantity  $f(P)$  by the computable plug-in estimate  $f(\hat{P}_n)$ .

Of course, not every functional behaves continuously with respect to the underlying distribution. The rigorous theory of the bootstrap studies conditions under which the approximation

$$f(\hat{P}_n) \approx f(P) \quad (21)$$

is valid. Roughly speaking, the bootstrap works well when the statistic of interest is a sufficiently smooth functional of the population distribution.

The Glivenko–Cantelli theorem therefore provides the first hint as to why the bootstrap might work: the empirical distribution converges to the truth, and many statistical quantities vary continuously with the underlying distribution.

## 2.4 When Can the Bootstrap Fail?

The bootstrap is based on the idea that the empirical distribution  $\hat{P}_n$  is close to the unknown population distribution  $P$ . However, closeness of distributions is not enough. The quantity being estimated must also depend smoothly on the underlying distribution.

A classical example where difficulties arise is the sample maximum

$$T_n = \max(X_1, \dots, X_n). \quad (22)$$

Suppose that

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, 1). \quad (23)$$

Let

$$M_n = \max(X_1, \dots, X_n) \quad (24)$$

denote the observed maximum.

A bootstrap sample is drawn from the empirical distribution  $\hat{P}_n$ , so every bootstrap observation must be one of the original data points. Consequently,

$$T_n^* = \max(X_1^*, \dots, X_n^*) \leq M_n. \quad (25)$$

In fact,  $T_n^* = M_n$  whenever the largest observed data point appears at least once in the bootstrap sample. Since each bootstrap draw selects this observation with probability  $(1/n)$ ,

$$P^*(T_n^* = M_n) = 1 - \left(1 - \frac{1}{n}\right)^n. \quad (26)$$

Therefore, as  $n \rightarrow \infty$ ,

$$P^*(T_n^* = M_n) \longrightarrow 1 - e^{-1} \approx 0.63. \quad (27)$$

Thus, approximately 63 percent of the mass of the bootstrap distribution of the sample maximum is at the single point  $M_n$ . This concentration of probability mass at a single point substantially reduces the variance of the bootstrap distribution.

As a result, the bootstrap variance

$$\text{Var}_*(T_n^*) \quad (28)$$

tends to underestimate the true sampling variance

$$\text{Var}(T_n). \tag{29}$$

The sample maximum therefore provides a classical example of a statistic for which the ordinary bootstrap fails.

More generally, bootstrap methods can perform poorly for statistics that depend heavily on extreme observations or boundary behavior. Such statistics are not sufficiently smooth functionals of the underlying distribution, and the plug-in principle underlying the bootstrap breaks down.

*Remark 1.* A central theme of modern bootstrap theory is that the bootstrap tends to work whenever the statistic of interest is a sufficiently smooth functional of the population distribution. This includes many commonly used estimators such as means, quantiles, regression coefficients. In more advanced treatments, this smoothness is formalized through notions such as influence functions and Hadamard differentiability. The sample maximum illustrates what can go wrong when this smoothness property fails.

## 2.5 Beyond Standard Errors

Throughout this chapter, we have focused on one particular application of the bootstrap: estimating the variance or standard error of a statistic. Historically, this was one of the original motivations for the method. However, the scope of the bootstrap is much broader.

Because the bootstrap approximates the entire sampling distribution of a statistic, the bootstrap can be used for a wide variety of inferential tasks, including estimating standard errors and bias, constructing confidence intervals, performing hypothesis tests, assessing the stability of statistical procedures, and quantifying uncertainty in machine learning algorithms. Over the years, many refinements and extensions of the basic bootstrap have been developed. Over the years, a number of important variants of the bootstrap have been developed, including the percentile bootstrap, the bias-corrected and accelerated (BCa) bootstrap, the Bayesian bootstrap, the parametric bootstrap, and block bootstrap methods for dependent data and time series. Today, bootstrap methods are among the most widely used tools in modern statistics and machine learning. Their appeal stems from a simple idea: replace an analytically intractable sampling problem by a computational one. As computing power has increased, the bootstrap has become a standard component of the statistician's toolkit. For a deeper treatment of bootstrap methods, see [Efron and Tibshirani \[1993\]](#), [Davison and Hinkley \[1997\]](#), and [Efron and Hastie \[2016\]](#).

## References

- Anthony C. Davison and David V. Hinkley. *Bootstrap Methods and Their Application*. Cambridge University Press, Cambridge, 1997.
- Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1): 1–26, 1979.
- Bradley Efron and Trevor Hastie. *Computer Age Statistical Inference: Algorithms, Evidence, and Data Science*. Cambridge University Press, Cambridge, 2016.
- Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, New York, 1993.